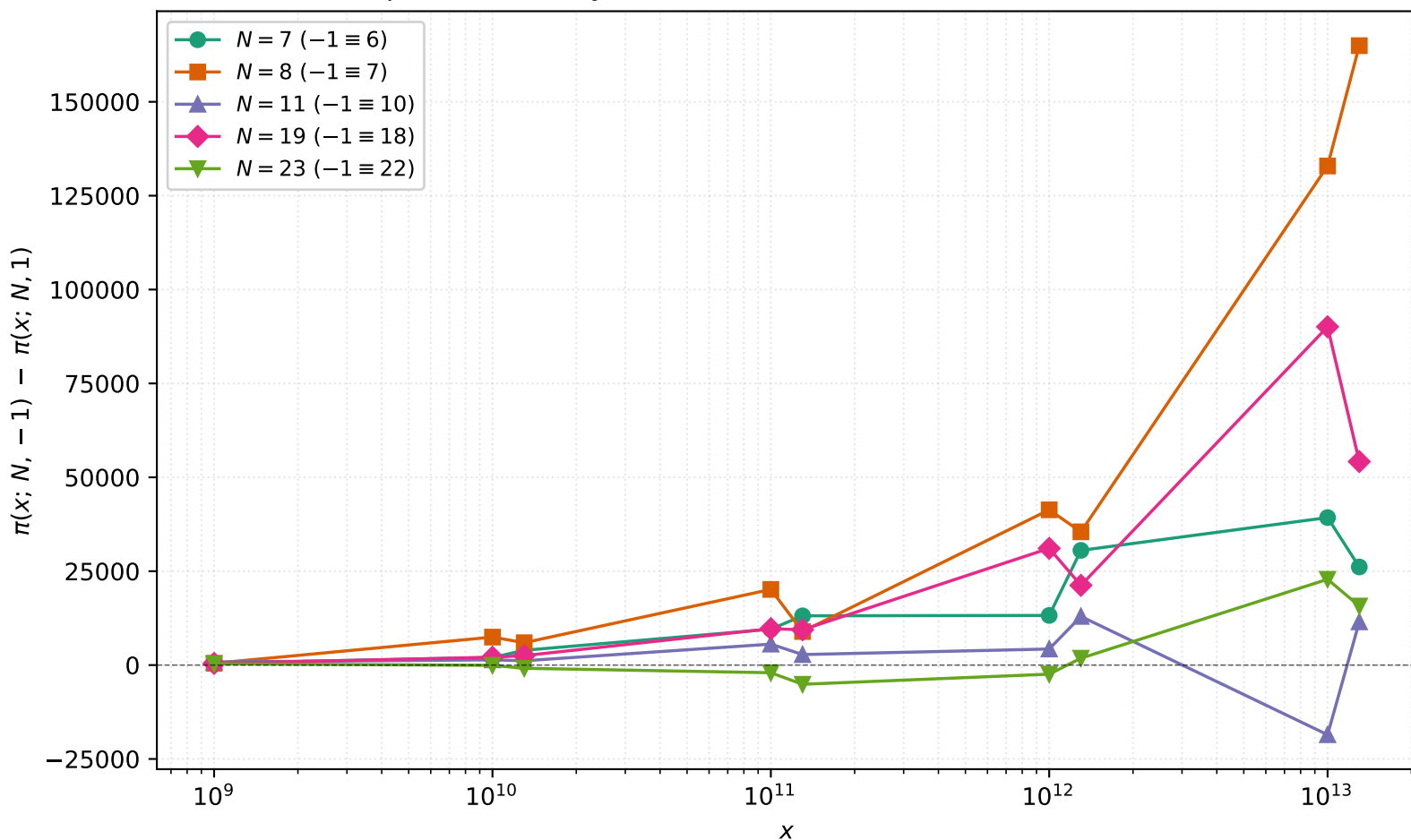


Replication of Koyama (2026): dominance of $-1 \pmod{N}$ vs x Rank of $-1 \pmod{N}$ among non-residues (smaller = stronger dominance)